

Critical Stokes Reversion and q -Gamma Cores

Exact Mellin–Borel completion,
 q -Pochhammer summability, and ramified inverse transseries

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Abstract

We develop a branch-resolved reversal calculus for sectorial complex transseries and apply it to a moving-parameter q -factorial and to the q -gamma function. An explicit Mellin–Borel identity reconstructs $\log(e^{-ah}; e^{-h})_\infty$ from its lateral Borel sums and convergent dual products. After a rotation of variables, this proves the formulas stated as Conjecture 5 in Fantini–Rella, arXiv:2506.08265v2. A finite Fourier–Möbius argument characterizes every rational-parameter weighted sum whose median recovers the original function; the Dirichlet-character case proves their Conjecture 1.

For inverse problems, a regular inverse transports a small jump by a reciprocal derivative, whereas an order- r critical point can change the action from A to A/r . We prove an all-orders analytic blow-up theorem, a uniform fold chart, and a jet-cancellation rule. For the natural minimum of $\log \Gamma_{e^{-h}}(a)$, an exact correction of size $e^{-4\pi^2/h}$ separates inverse roots on the scale $e^{-2\pi^2/h}$. We compute the coefficient, the moving critical value, and the first three ramified sectors. We also reverse the q -factorial with respect to h , where its dual action becomes an action-24 sector in the dominant inverse coordinate.

Conventional proofs and reproducible exact and high-precision checks are provided. Stokes discontinuities, independent flat completion data, logarithmic monodromy, and inverse-sheet branching are kept distinct.

Research status. Theorems are proved in the specified local domains. The new proof and applications have not been independently refereed or formalized in a proof assistant. First-priority claims are not made; the comparison is with the inspected sources, not an exhaustive literature or repository audit.

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1 The questions answered and the source boundary

1.1 Why a new inverse chart is necessary

The relevant difficulty is not the existence of the formal Lagrange–Bürmann formula. It is deciding which analytic function is being inverted, on which sheet, and to what absolute accuracy. Three operations that look similar in a purely symbolic calculation have different meanings:

formal reversion, sectorial inversion, physical-function inversion.

The first need not determine the third. A prescribed summation convention can still miss a nonzero function flat at the expansion point. Near a critical value this missing term can dominate the inverse error.

Our principal special-function variables are

$$h = -\operatorname{Log} q, \quad \Re h > 0, \quad A = 4\pi^2, \quad Q = e^{-A/h}, \quad (1.1)$$

with $h \rightarrow 0$ on closed subsectors of the right half-plane. The parameter a is initially real, $0 < a < 1$. We later continue locally to complex neighborhoods of every positive real a . The moving parameter $z = q^a = e^{-ah}$ tends to 1, precisely where a fixed- z polylogarithmic expansion ceases to be uniform.

1.2 Relation to the ProveIt program

The repository documentation was inspected at commit

`b484725c893a7a9bde856151988996aef6965cb0`.

The canonical transseries volume and its documentation distinguish formal inversion, dominant cores, and residual certificates; the companion volume treats combinatorial inverses and q -products [1]. The preceding report *Complex Transseries Reversion at q -Cusps* treats finite-action convergent units and modular Euler quotients. Its fixed-parameter shifted-factorial inverse is explicitly asymptotic rather than identified with a Borel sum. Moving gamma cores and summation-compatible inversion are listed there as further questions [2].

We do not repeat its eta-quotient classification, cusp-data recovery, or four-factor flatness construction. Instead, we provide a divergent algebraic block with a fully identified analytic completion, then reverse it in both a regular and a critical chart. The present article is self-contained: no mathematical claim here depends on an unchecked statement in a repository draft. The repository’s scoped Lean results do not imply formal verification of the new theorems.

1.3 Relation to published work

Fantini–Rella compute the rational-parameter Borel singularities and Stokes constants of q -Pochhammer expansions. In the inspected version, arXiv:2506.08265v2, dated 1 April 2026, their Conjecture 5 identifies certain lateral sums with corrected products; their Conjecture 1 concerns median reconstruction of character-weighted sums [3]. Section 6 gives a proof of that completion identity, and Section 7 proves a stronger arbitrary-weight criterion. The source’s general modular-resurgence conjectures are *not* thereby resolved.

The Bernoulli expansions are classical; we credit McIntosh and the recent moving-parameter treatment of Arabi Ardehali–Rosengren [5, 4]. Nonlinear stability of appropriate resurgent and Borel-summable classes under inversion is also established mathematics [6, 7]. Our proof does

not present that closure theorem as new, and does not infer analytic summability solely from formal closure. The elementary Mellin identity below supplies the exact completion needed for this particular family.

1.4 Main result ledger

- R1.** An explicit, normally convergent meromorphic Borel kernel, including an entire-in- a representation suitable for parameter differentiation, and an exact lateral completion formula.
- R2.** A proof of the two specified Fantini–Rella conjectures, with the rotation, branch convention, and range $1 \leq k < N$ written explicitly. The weighted criterion does not require multiplicativity or primitivity.
- R3.** A local critical reversion theorem: an action A becomes A/r under a nondegenerate order- r balance. For folds, a moving-critical-value chart gives all orders without a singular division by the forward derivative.
- R4.** Exact q -gamma completion and ramified inverse sectors near its natural minimum, together with regular inverses in a and in h .
- R5.** Counterexamples to two tempting shortcuts: zero Stokes jump does not imply zero flat correction, and arithmetic averaging does not commute with nonlinear inversion.

2 Complex transseries require charts and realizations

2.1 Sectorial scales

On a simply connected sector

$$S = \{0 < |h| < h_0 : \theta_0 < \arg h < \theta_1\},$$

choose branches of $\text{Log } h$ and of all algebraic powers. Expressions such as

$$\sum_{\alpha \in \Lambda} e^{-\alpha/h} h^{\beta_\alpha} (\text{Log } h)^{m_\alpha} \hat{f}_\alpha(h) \tag{2.1}$$

become useful only after specifying the support convention and the meaning of each coefficient block $\hat{f}_\alpha$. A closed subsector is *small for an action* α when $\Re(\alpha/h) > c/|h|$ there for some $c > 0$.

Finite collections of small actions generate locally finite positive projection budgets. Resonant actions must be combined. At a direction where $\Re((\alpha - \beta)/h) = 0$, two exponential magnitudes tie; that is a *dominance wall*, not by itself a Borel-summation discontinuity. A Stokes direction requires a specified Borel singularity and a specified summation problem. For the present tail, the positive real h direction is a Stokes direction, while $\Re(A/h) = 0$ lies on the boundary of the right half-plane.

2.2 A formal contraction principle

Lemma 2.1 (Filtered implicit inversion). *Let $\mathcal{A}$ be a complete separated algebra over $\mathbb{C}$, with a descending multiplicative filtration I^n . Suppose $T : I \rightarrow I$ is a well-defined map satisfying*

$$u - v \in I^n \implies T(u) - T(v) \in I^{n+1}.$$

Then $u = T(u)$ has exactly one solution in I . Picard iteration starting at 0 computes it to every filtration order.

Proof. The consecutive iterates differ in I^n after n steps, hence converge in the complete topology. The contraction property makes T continuous and the limit fixed. Two fixed points have difference in every I^n , and separatedness makes them equal. $\square$

This elementary lemma is deliberately not a universal transseries-field construction. Taylor substitution, differentiation, and action composition must first be defined in the algebra being used.

For a regular implicit equation $F_0(x) + R(x, \mathbf{t}) = y$, with $F'_0(x_0) \neq 0$, the correction $x - x_0$ has this form in the ideal generated by the flat parameters $\mathbf{t}$. At a critical point the linear coefficient is not a unit. A ramified substitution is required before applying the same principle.

2.3 Analytic realizations and nonlinear closure

We use two distinct analytic settings. In the first, the algebraic blocks are actual sectorial holomorphic functions and the dependence on independent flat parameters is convergent. Ordinary analytic implicit inversion is sufficient. In the second, blocks are Gevrey series with specified Borel transforms and exponential bounds. Lateral summation then produces the actual functions to which the first setting applies.

In a class for which Borel–Laplace summation is an algebra morphism and is compatible with analytic substitution and the implicit-function operation, uniqueness gives

$$\mathcal{S}_\theta(\widehat{G}) = (\mathcal{S}_\theta(\widehat{F}))^{-1} \quad (2.2)$$

on matching regular inverse charts. The established closure results in [6] justify this procedure under their hypotheses. Equation (2.2) does not assert that the right side is the inverse of an arbitrarily chosen special function having that expansion. Our completion theorem determines the additional terms explicitly.

2.4 A countable analytic version

Proposition 2.2 (Normally summable perturbations). *Let F_0 be holomorphic on $|x - x_0| < R$, with $F'_0(x_0) = b \neq 0$. Suppose d_j are bounded holomorphic functions on that disk and $\sum_j |t_j| \|d_j\|_R < \infty$. For this sum sufficiently small,*

$$F_0(x) + \sum_j t_j d_j(x) = F_0(x_0)$$

has one root in a fixed smaller disk, near x_0 . That root is a holomorphic function of the weighted ℓ^1 parameter $(t_j \|d_j\|_R)_j$. Its expansion has the usual convergent homogeneous multilinear terms.

Proof. Normal convergence gives a bounded holomorphic perturbation. Cauchy’s estimate bounds its derivative on a smaller disk by a constant times the weighted norm. Shrink that disk until F'_0 is close to b ; then shrink the parameter ball until the map $x \mapsto x - [F_0(x) + \sum t_j d_j(x) - F_0(x_0)]/b$ is a uniform contraction mapping the disk into itself. Iterates are holomorphic in the Banach parameter; locally uniform convergence gives a holomorphic fixed point. Cauchy estimates on the parameter ball give its Taylor expansion. $\square$

Thus analytic inversion can accommodate even accumulating action values when their amplitudes are normally summable. This does not automatically turn the resulting expression into a formally locally finite action expansion. Analytic summability and formal support admissibility are different requirements.

3 Regular inversion, jump transport, and error certificates

3.1 An all-orders perturbation formula

Theorem 3.1 (Regular inverse response). *Let f be univalent near x_0 , write $g = f^{-1}$, and let d be holomorphic there. For sufficiently small independent t , the local inverse g_t of $f_t = f + td$ is holomorphic in (y, t) and*

$$g_t(y) = g(y) + \sum_{n \geq 1} \frac{(-t)^n}{n!} \frac{d^{n-1}}{dy^{n-1}} [g'(y)d(g(y))^n]. \quad (3.1)$$

The series converges normally on smaller product neighborhoods.

Proof. Put $z = f(x)$, so that $z + tD(z) = y$ with $D = d \circ g$. Classical Lagrange inversion with the observable $g(z)$ gives (3.1). Alternatively, analytic implicit inversion supplies the convergent solution, and differentiating its defining equation computes the same coefficients uniquely. This is the classical observable form of Lagrange–Bürmann, used here with an independent flat parameter rather than a numerical perturbation of unspecified size. $\square$

In source coordinates, the first terms are

$$g_t - g = -t \frac{d}{f'} + t^2 \left(\frac{dd'}{(f')^2} - \frac{f''d^2}{2(f')^3} \right) + O(t^3), \quad (3.2)$$

where the right side is evaluated at $x = g(y)$. For a perturbation $d(x, t)$ holomorphic in t , the analytic implicit theorem still gives all orders; the coefficients are obtained by expanding the defining equation, or by introducing an auxiliary multiplier before using (3.1).

3.2 Source and target Stokes maps

Suppose two sectorial functions F_- and F_+ are locally univalent and have inverses G_- and G_+ on the selected charts. There are two useful, but different, transition conventions:

$$F_+ = T \circ F_- \quad \implies \quad G_+ = G_- \circ T^{-1}, \quad (3.3)$$

$$F_+ = F_- \circ U \quad \implies \quad G_+ = U^{-1} \circ G_-. \quad (3.4)$$

These identities are exact. For an additive forward jump $F_+ = F_- + \Delta$, Theorem 3.1 gives its inverse expansion by placing an auxiliary parameter in front of Δ .

On triple overlaps, transition maps compose in their actual source or target order. Taking inverses reverses that order. No new Stokes constant is determined from formal coefficients alone by these identities; one must first identify the forward sectorial functions and their jumps.

3.3 An inverse-existence certificate

Lemma 3.2 (Disk certificate). *Let f be holomorphic on a neighborhood of the closed disk $D = \{|x - z| \leq r\}$. Suppose $b \neq 0$, $0 \leq \kappa < 1$, and*

$$\sup_D |1 - f'(x)/b| \leq \kappa, \quad |f(z) - y|/|b| \leq (1 - \kappa)r.$$

Then $f(x) = y$ has exactly one root in D , and

$$|x - z| \leq \frac{|f(z) - y|}{(1 - \kappa)|b|}. \quad (3.5)$$

Proof. The map $T(x) = x - (f(x) - y)/b$ is κ -Lipschitz on the convex disk, by integrating its derivative along segments. The residual bound makes $T(D) \subseteq D$. The contraction theorem gives the root and uniqueness; $|x - z| \leq \kappa|x - z| + |T(z) - z|$ gives the estimate. $\square$

A bare lower bound for $|f'|$ on an arbitrary complex domain would not justify the segment estimate. The explicit closeness to one constant b is an essential feature of this certificate.

3.4 Averaging is not a nonlinear summation rule

Proposition 3.3 (Median and inverse do not commute). *Let $f_{\pm} = f \pm td/2$, and let $g_{\pm} = f_{\pm}^{-1}$. Then*

$$\frac{g_+ + g_-}{2} = g + \frac{t^2}{8} \frac{d}{dy} \left[g'(y) d(g(y))^2 \right] + O(t^4). \quad (3.6)$$

Thus the arithmetic mean of inverse branches is generally not the inverse of the arithmetic mean of the forward branches.

Proof. Substitute $t/2$ and $-t/2$ into (3.1) and average. All odd powers cancel; the displayed quadratic term remains. $\square$

We use $\mathcal{M} = (\mathcal{S}_+ + \mathcal{S}_-)/2$ only for the linear logarithmic Borel problem below. An algebra-compatible *Écalte median* on a full transseries space is a different construction. It must not be replaced by arithmetic averaging after nonlinear operations.

4 Critical reversion and fractional actions

4.1 Uniform blow-up at an order- r critical point

Theorem 4.1 (Critical flat-parameter reversion). *Let $c(h)$ be holomorphic on a small punctured h -sector, and let $F(x, h, t)$ be jointly holomorphic on moving neighborhoods of $(c(h), h, 0)$, with uniform x, t radii and bounds. Assume*

$$F(c + z, h, 0) = v(h) + a_r(h)z^r + O(z^{r+1}), \quad r \geq 2, \quad (4.1)$$

$$F(c + z, h, t) = F(c + z, h, 0) + t d(c + z, h) + O(t^2), \quad |a_r(h)| \geq a_0 > 0. \quad (4.2)$$

Fix compact simply connected target patches avoiding $Y = d(c(h), h)$ uniformly. On chosen root branches, the equation $F(x, h, t) = v(h) + tY$ has r local solutions, holomorphic in $s = t^{1/r}$, of the form

$$x_j = c(h) + s \omega_j \left(\frac{Y - d(c(h), h)}{a_r(h)} \right)^{1/r} + O(s^2), \quad \omega_j^r = 1. \quad (4.3)$$

Each branch has a convergent expansion in s , locally uniformly in Y, h . Truncating after s^M has error $O(|s|^{M+1})$.

Proof. Set $x = c + su$, $t = s^r$, subtract $v + s^r Y$, and divide by s^r . The result extends holomorphically to $s = 0$ and is

$$a_r u^r + d(c, h) - Y + s R(u, s, Y, h) = 0.$$

Its r limiting roots are nonzero and separated on the stated compact patches. Their u -derivatives, $ra_r u^{r-1}$, stay away from zero. The analytic implicit theorem gives one branch near each limiting root, with uniform radii after shrinking the patches. These account for all local roots in the corresponding scaled neighborhoods, by Rouché's theorem. Cauchy's estimate in s gives the remainder. $\square$

For $t = e^{-A/h}$, choose $s = e^{-A/(rh)}$ on the sector. This proves an action-division law rather than merely a formal square-root analogy. It requires the nonvanishing leading balance. The excluded target value is where another, finer critical scaling is necessary.

Comparing the unperturbed and perturbed inverse branches at the same $y = v + tY$, on patches avoiding both 0 and $d(c, h)$, gives

$$G_{t,j} - G_{0,j} = s \omega_j a_r^{-1/r} \left((Y - d(c, h))^{1/r} - Y^{1/r} \right) + O(s^2). \quad (4.4)$$

At such targets, linearizing a jump as $-\Delta/F'$ is not uniformly valid: the nonlinear root difference is of the same order as each root displacement.

4.2 An exact moving-fold chart

Theorem 4.2 (Fold chart through the moving critical value). *Under (4.1)–(4.2) with $r = 2$, there are unique holomorphic functions $c_t(h)$ and $v_t(h)$ near $c(h), v(h)$ such that $F_x(c_t, h, t) = 0$ and $v_t = F(c_t, h, t)$. There is a holomorphic local coordinate $w = w(x, h, t)$ with nonzero x -derivative for which*

$$F(x, h, t) = v_t(h) + w(x, h, t)^2. \quad (4.5)$$

Consequently, on a chosen square-root cover, both inverse sheets are

$$x = \chi(\pm \sqrt{y - v_t(h)}, h, t), \quad (4.6)$$

where χ is holomorphic. This representation remains valid at the moving branch point; only the square root is ramified.

Proof. Apply the analytic implicit theorem to $F_x = 0$, using $F_{xx}(c, h, 0) \neq 0$. Taylor's formula factors $F(x, h, t) - v_t = (x - c_t)^2 A(x, h, t)$ with A nonvanishing near the base point. Choose one analytic square root of A , and put $w = (x - c_t)\sqrt{A}$. Its derivative at c_t is nonzero, so it has a holomorphic local inverse χ . Uniformity follows on the same smaller parameter neighborhoods as before. $\square$

This is the one-variable parametric Morse construction, not a new singularity-classification theorem. Its role is to provide a correctly centered inverse-transseries chart.

Writing $F_0 = F(\cdot, h, 0)$, the first moving-center terms are

$$c_t = c - t \frac{d'(c)}{F_0''(c)} + O(t^2), \quad (4.7)$$

$$v_t = v + td(c) + t^2 \left(e(c) - \frac{d'(c)^2}{2F_0''(c)} \right) + O(t^3), \quad (4.8)$$

when $F = F_0 + td + t^2e + O(t^3)$. All coefficients retain their full h dependence.

4.3 Three explicit ramified sectors

At a fold, write

$$\begin{aligned} F_0(c + z) &= v + \alpha_2 z^2 + \alpha_3 z^3 + \alpha_4 z^4 + O(z^5), \\ d(c + z) &= d_0 + d_1 z + d_2 z^2 + O(z^3), \quad e(c) = e_0. \end{aligned}$$

For $y = v + tY$, put $s = \sqrt{t}$ and $x = c + u_0s + u_1s^2 + u_2s^3 + O(s^4)$. Direct coefficient matching gives

$$u_0^2 = \frac{Y - d_0}{\alpha_2}, \quad (4.9)$$

$$u_1 = -\frac{\alpha_3 u_0^2 + d_1}{2\alpha_2}, \quad (4.10)$$

$$u_2 = -\frac{\alpha_2 u_1^2 + 3\alpha_3 u_0^2 u_1 + \alpha_4 u_0^4 + d_1 u_1 + d_2 u_0^2 + e_0}{2\alpha_2 u_0}. \quad (4.11)$$

The nonzero denominator in the last equation is exactly the simple-root condition in the scaled chart. This recurrence continues to all orders, and Theorem 4.1 proves convergence, not just consistency of a formal calculation.

4.4 Jet cancellation changes the action rule

Proposition 4.3 (Shared-factor cancellation). *Suppose at the fixed target v an analytic family factors exactly as*

$$F(c + z, t) - v = z^m (z^{r-m} A(z) + tD(z, t)), \quad 0 \leq m < r,$$

with $A(0)D(0,0) \neq 0$. There are m roots pinned at c , counted with multiplicity, and $r - m$ moving roots with scale $t^{1/(r-m)}$ and convergent ramified expansions.

Proof. The first factor supplies the pinned roots. Apply Theorem 4.1, or its proof, to the second factor after $z = t^{1/(r-m)}u$. $\square$

For a fold with a perturbation vanishing at c but not in its first derivative, the moving-root scale is t , not $\sqrt{t}$. The critical value first changes at order t^2 . Vanishing of one leading coefficient must not be confused with the exact factorization used in this proposition.

For several commensurate actions, the lower Newton polygon of the monomials $t^n z^m$ organizes these balances. This is the usual Newton–Puiseux mechanism applied to a flat parameter. Arbitrary incommensurate or accumulating action sets require additional support and convergence hypotheses; no global Newton–Puiseux theorem for all formal complex transseries is claimed here.

5 The moving q -factorial and its Borel transform

5.1 Forward expansion and conventions

Define

$$p(a, h) = \log(e^{-ah}; e^{-h})_\infty = -\sum_{n \geq 1} \frac{e^{-ahn}}{n(1 - e^{-hn})}. \quad (5.1)$$

For real $a > 0$ and $\Re h > 0$, the right side fixes the analytic logarithm. The Bernoulli polynomials are defined by

$$\frac{se^{as}}{e^s - 1} = \sum_{n \geq 0} B_n(a) \frac{s^n}{n!}, \quad B_n = B_n(0), \quad B_1 = -\frac{1}{2}.$$

The elementary-gamma core and the formal tail are

$$E(a, h) = -\frac{\pi^2}{6h} + \left(\frac{1}{2} - a\right) \text{Log } h + \frac{1}{2} \log(2\pi) - \log \Gamma(a) + \frac{B_2(a)}{4} h, \quad (5.2)$$

$$\hat{\Phi}_a(h) = \sum_{k \geq 1} \frac{B_{2k} B_{2k+1}(a)}{2k(2k+1)!} h^{2k}, \quad \hat{p} = E - \hat{\Phi}_a. \quad (5.3)$$

This is the moving-parameter gamma expansion in the classical q -factorial calculus [5, 4]; we derive the exact remainder rather than assume it.

Our Borel–Laplace normalization is

$$\mathcal{B}(h^n) = \frac{\xi^{n-1}}{(n-1)!}, \quad n \geq 1, \quad \mathcal{L}_\theta b(h) = \int_0^{e^{i\theta}\infty} e^{-\xi/h} b(\xi) d\xi. \quad (5.4)$$

There is no extra $1/h$ factor. The operators $\mathcal{S}_+$ and $\mathcal{S}_-$ refer, respectively, to contours above and below the positive real Borel poles. The core E is carried along exactly, with its chosen logarithm.

5.2 A parameter-holomorphic kernel

Theorem 5.1 (Explicit Borel kernel). *The Borel transform $b_a = \mathcal{B}\hat{\Phi}_a$ is the germ at zero of*

$$b_a(\xi) = -\frac{2}{\xi} \sum_{n \geq 1} \left\{ \frac{\sin((a - \frac{1}{2})\xi/(2\pi n))}{2 \sin(\xi/(4\pi n))} - \left(a - \frac{1}{2}\right) \right\}. \quad (5.5)$$

The apparent singularity at zero is removable. The sum converges normally on compact subsets away from $\xi \in 4\pi^2\mathbb{Z} \setminus \{0\}$, uniformly on compact subsets of the entire a -plane. Its only possible nonzero Borel singularities are simple poles at those locations. For $N \geq 1$,

$$\text{Res}_{\xi=4\pi^2 N} b_a(\xi) = \text{Res}_{\xi=-4\pi^2 N} b_a(\xi) = -\frac{1}{\pi} \sum_{d|N} \frac{\sin(2\pi da)}{d}. \quad (5.6)$$

Proof. Set

$$T_a(s) = \frac{\sinh((a - \frac{1}{2})s)}{2 \sinh(s/2)} - \left(a - \frac{1}{2}\right) = \sum_{k \geq 1} \frac{B_{2k+1}(a)}{(2k+1)!} s^{2k}. \quad (5.7)$$

The last identity follows by taking the odd part of the Bernoulli polynomial generating function and dividing by s . Since

$$\frac{B_{2k}}{(2k)!} = \frac{2(-1)^{k+1}\zeta(2k)}{(2\pi)^{2k}},$$

Borel transformation of (5.3) gives $-2\xi^{-1} \sum_{n \geq 1} T_a(i\xi/(2\pi n))$, which is (5.5). On compact parameter sets the summand is $O(|\xi|^2/n^2)$ for large n . This proves normal convergence and removability at zero.

At $\xi = 4\pi^2 N$, only the finitely many terms with $n \mid N$ have poles. Put $d = N/n$ and use $\sin(2\pi d(a - \frac{1}{2})) = (-1)^d \sin(2\pi da)$. The residue of the corresponding term is $-\sin(2\pi da)/(\pi d)$. Summation gives the residue at the positive pole. The kernel is odd in ξ , so residues at opposite poles are equal. $\square$

The rational-parameter singularities and their divisor-sum Stokes data are already present in [3]. Representation (5.5) is useful here because it remains valid when a is differentiated or moved off the real axis. A Fourier series with coefficients $\sin(2\pi ma)$ generally diverges when $\Im a \neq 0$; we do not use such a series to claim complex-parameter uniformity.

5.3 The real-parameter Fourier kernel and its bounds

Lemma 5.2. *For real $0 < a < 1$, one also has*

$$b_a(\xi) = \frac{2}{\pi} \sum_{m,n \geq 1} \frac{\sin(2\pi ma)}{m} \frac{\xi}{(4\pi^2 mn)^2 - \xi^2}. \quad (5.8)$$

On each ray whose angle is bounded away from 0 and π , this kernel is bounded at infinity and is $O(\xi)$ at zero, uniformly for real $a \in [0, 1]$. The double series can be integrated termwise in a convergent Laplace direction.

Proof. Use the Fourier expansion

$$B_{2k+1}(a) = (-1)^{k+1} \frac{2(2k+1)!}{(2\pi)^{2k+1}} \sum_{m \geq 1} \frac{\sin(2\pi ma)}{m^{2k+1}}, \quad k \geq 1.$$

Together with the Bernoulli-number identity in the preceding proof, this sums a geometric series in ξ^2 and yields (5.8), initially near zero. Both sides continue meromorphically, and the double series is normally convergent on compact sets away from its poles.

If $r = |\xi|$ lies on an allowed ray, then

$$\left| \frac{\xi}{(4\pi^2 mn)^2 - \xi^2} \right| \leq C_\theta \frac{r}{(mn)^2 + r^2}.$$

For $m \leq r$ the sum over n is $O(1/m)$, and for $m > r$ it is $O(r/m^2)$. The additional factor $1/m$ makes the sums over m bounded. Near zero use $\sum m^{-3}n^{-2} < \infty$. These bounds, multiplied by the decaying Laplace exponential, justify Fubini's theorem. $\square$

Consequently, the tail is Gevrey 1 and summable away from the two real Borel rays. For complex a in a fixed small neighborhood, the parameter-holomorphic kernel gives exponential-type bounds on such rays; sufficiently small h then suffices. Integer shifts in a can also be handled by the exact q -factorial recurrence.

6 An exact Mellin–Borel completion theorem

6.1 The completion identity

Define the convergent dual logarithms by their zero-constant-term series:

$$L_\pm(a, Q) = \log(e^{\pm 2\pi i a} Q; Q)_\infty = - \sum_{m \geq 1} \frac{e^{\pm 2\pi i a}}{m} \frac{Q^m}{1 - Q^m}. \quad (6.1)$$

For real a they are holomorphic for $|Q| < 1$. Locally in complex a , the same is true whenever $|Q|e^{2\pi|\Im a|} < 1$.

Theorem 6.1 (Exact lateral reconstruction). *For $0 < a < 1$ real and $\Re h > 0$, with the conventions above,*

$$\mathcal{S}_+ \hat{p}(a, h) = p(a, h) - L_-(a, Q), \quad (6.2)$$

$$\mathcal{S}_- \hat{p}(a, h) = p(a, h) - L_+(a, Q). \quad (6.3)$$

In particular,

$$p(a, h) = E(a, h) - \mathcal{M} \hat{\Phi}_a(h) + \frac{L_+(a, Q) + L_-(a, Q)}{2}, \quad \mathcal{M} = \frac{\mathcal{S}_+ + \mathcal{S}_-}{2}, \quad (6.4)$$

and the tail has the exact Stokes jump

$$\begin{aligned} \mathcal{S}_+ \widehat{\Phi}_a - \mathcal{S}_- \widehat{\Phi}_a &= L_-(a, Q) - L_+(a, Q) \\ &= 2i \sum_{N \geq 1} \left(\sum_{d|N} \frac{\sin(2\pi da)}{d} \right) Q^N. \end{aligned} \quad (6.5)$$

These identities continue locally in a near positive real parameters, for sufficiently small h on the corresponding sectorial domains.

The main point is the *even* combination $(L_+ + L_-)/2$, which is not determined by the discontinuity $L_- - L_+$. Computing residues alone would prove the jump but would not prove reconstruction of the physical product. We therefore prove (6.4) independently by Mellin inversion.

6.2 A principal-value kernel

Lemma 6.2 (Mellin transform of the symmetric kernel). *For $T > 0$, let*

$$K(T) = \text{PV} \int_0^\infty e^{-Tt} \frac{t}{1-t^2} dt = \frac{1}{2} \left(e^{-T} \text{Ei}(T) - e^T E_1(T) \right). \quad (6.6)$$

Then $K(T) = O(1 + |\log T|)$ at zero and $K(T) = O(T^{-2})$ at infinity. For $0 < \Re u < 2$,

$$\int_0^\infty T^{u-1} K(T) dT = -\frac{\pi}{2} \Gamma(u) \cot \frac{\pi u}{2}. \quad (6.7)$$

Consequently, for $1 < c < 2$,

$$\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \Gamma(u) \cot \frac{\pi u}{2} T^{-u} du = -\frac{2}{\pi} K(T). \quad (6.8)$$

Proof. The partial fraction decomposition $t/(1-t^2) = \frac{1}{2}((1-t)^{-1} - (1+t)^{-1})$ gives the exponential-integral expression, with a symmetric principal value at $t = 1$. It also gives the asserted endpoint estimates, or they follow by splitting the integral near $0, 1, \infty$.

For rigor in exchanging Mellin integration and principal value, first exclude $(1-\delta, 1+\delta)$. Away from that interval Fubini applies, since the absolute t -integral of $t^{1-\Re u}/|1-t^2|$ is finite at both endpoints when $0 < \Re u < 2$. The difference between this truncated kernel and its principal value is bounded by $C\delta(1+T)e^{-T/2}$ for $0 < \delta < 1/2$, after subtracting the pole numerator's value at $t = 1$. This bound is integrable against $T^{\Re u-1} dT$. Hence the limit commutes with Mellin integration. The transform is

$$\Gamma(u) \text{PV} \int_0^\infty \frac{t^{1-u}}{1-t^2} dt = \frac{\Gamma(u)}{2} \text{PV} \int_0^\infty \frac{v^{-u/2}}{1-v} dv = -\frac{\pi}{2} \Gamma(u) \cot \frac{\pi u}{2}.$$

The last principal-value identity follows, for example, by splitting at 1, substituting $v \mapsto 1/v$ in the second piece, and applying the partial fraction expansion of the cotangent. Mellin inversion is valid in this strip; its right side has exponential decay on vertical lines from $\Gamma(u)$, while the cotangent is bounded there. This proves (6.8). $\square$

6.3 Mellin representation of the product

Lemma 6.3. *For $h > 0$, $a > 0$, and $c_0 > 1$,*

$$p(a, h) = -\frac{1}{2\pi i} \int_{c_0-i\infty}^{c_0+i\infty} \Gamma(s) \zeta(s+1) \zeta(s, a) h^{-s} ds. \quad (6.9)$$

For $0 < a < 1$, shift the contour to $-2 < \Re s < -1$. The residues crossed are exactly $E(a, h)$, so that

$$p(a, h) - E(a, h) = -\frac{1}{2\pi i} \int_{-c-i\infty}^{-c+i\infty} \Gamma(s) \zeta(s+1) \zeta(s, a) h^{-s} ds, \quad 1 < c < 2. \quad (6.10)$$

Proof. Expand the logarithm as a double sum $-\sum_{n \geq 1, j \geq 0} e^{-hn(j+a)}/n$ and insert Mellin inversion of the exponential. Absolute convergence for $\Re s > 1$ yields (6.9). Standard vertical-strip bounds for the zeta functions and exponential decay of Γ justify the contour shift. The pole at $s = 1$ contributes $-\pi^2/(6h)$. At $s = 0$, use

$$\zeta(0, a) = \frac{1}{2} - a, \quad \zeta'(0, a) = \log \Gamma(a) - \frac{1}{2} \log(2\pi),$$

and $\Gamma(s)\zeta(1+s) = s^{-2} + O(1)$. The residue, with the outer minus sign, is $(\frac{1}{2} - a) \log h - \log \Gamma(a) + \frac{1}{2} \log(2\pi)$. At $s = -1$, the residue is $B_2(a)h/4$ after the outer minus sign. These standard zeta identities are recorded in [8]. $\square$

6.4 Functional equations split the remainder

For real $0 < a < 1$, set

$$C_u(a) = \sum_{m \geq 1} \frac{\cos(2\pi ma)}{m^u}, \quad S_u(a) = \sum_{m \geq 1} \frac{\sin(2\pi ma)}{m^u}.$$

On the line in (6.10), the Hurwitz Fourier formula and the Riemann functional equation give

$$\zeta(s, a) = 2\Gamma(1-s)(2\pi)^{s-1} \left(\sin \frac{\pi s}{2} C_{1-s}(a) + \cos \frac{\pi s}{2} S_{1-s}(a) \right), \quad (6.11)$$

$$\zeta(s+1) = 2^{s+1} \pi^s \cos \frac{\pi s}{2} \Gamma(-s) \zeta(-s). \quad (6.12)$$

The first is DLMF 25.11.9 in changed variables [8]; the second is the standard functional equation [9]. Using $\Gamma(s)\Gamma(1-s) = \pi/\sin(\pi s)$ simplifies their product to the particularly useful identity

$$\Gamma(s)\zeta(s+1)\zeta(s, a) = (4\pi^2)^s \Gamma(-s) \zeta(-s) \left(C_{1-s}(a) + \cot \frac{\pi s}{2} S_{1-s}(a) \right). \quad (6.13)$$

Put $X = 4\pi^2/h$ and $u = -s$ in (6.10). Then

$$p - E = -\frac{1}{2\pi i} \int_{(c)} \Gamma(u) \zeta(u) \left(C_{1+u}(a) - \cot \frac{\pi u}{2} S_{1+u}(a) \right) X^{-u} du, \quad 1 < c < 2. \quad (6.14)$$

All Dirichlet series here converge absolutely, and all vertical integrals converge absolutely. We may therefore integrate the two double series separately. Mellin inversion of $\Gamma(u)$ evaluates the cosine part as

$$-\sum_{m, n \geq 1} \frac{\cos(2\pi ma)}{m} e^{-Xmn} = \frac{L_+(a, Q) + L_-(a, Q)}{2}. \quad (6.15)$$

Lemma 6.2 evaluates the sine part as

$$-\frac{2}{\pi} \sum_{m, n \geq 1} \frac{\sin(2\pi ma)}{m} K(Xmn). \quad (6.16)$$

The series in (6.16) converges absolutely: for fixed $X > 0$, its tail is bounded by a constant times $\sum m^{-3}n^{-2}$.

6.5 Identification with the lateral Borel sums

For a single pole pair, contour deformation above or below the positive pole gives

$$\int_0^{e^{\pm i\theta}\infty} e^{-\xi/h} \frac{\xi}{A_0^2 - \xi^2} d\xi = K(A_0/h) \pm \frac{\pi i}{2} e^{-A_0/h}, \quad h, A_0 > 0, \quad (6.17)$$

where $0 < \theta < \pi/2$, and deformation is within the indicated half-plane. The positive-pole residue is $-1/2$. By Lemma 5.2, we can sum the Laplace integrals termwise. The resulting series of K terms and residue exponentials are absolutely convergent. Therefore

$$\mathcal{M}\hat{\Phi}_a(h) = \frac{2}{\pi} \sum_{m,n \geq 1} \frac{\sin(2\pi ma)}{m} K(Xmn). \quad (6.18)$$

Equations (6.14)–(6.18) prove (6.4) for positive h .

The difference of the two contours is clockwise around each positive pole, so its residue contribution is $-2\pi i$ times the residue. Together with (5.6), this proves (6.5), including its sign. Substitution into the median identity gives (6.2)–(6.3).

For completeness, the lateral sums define holomorphic functions for $\Re h > 0$: choose a small positive, respectively negative, Borel ray whose Laplace exponential decays, and deform it within its half-plane. On overlapping h -domains the contours give the same analytic function. Lemma 5.2 supplies the required bounds. The right-hand sides of (6.2)–(6.3) are also holomorphic there. The identity theorem extends the equality from $h > 0$ to that entire half-plane.

Finally, (5.5) is normally convergent in complex a . Local exponential bounds allow differentiation under the Laplace integral for small h . Analytic continuation in a , followed when necessary by

$$p(a+1, h) = p(a, h) - \log(1 - e^{-ah}), \quad (6.19)$$

extends the identities to the stated positive-parameter neighborhoods. This finishes the proof of Theorem 6.1. $\square$

7 The conjecture comparison and a complete weight criterion

7.1 Exact translation of Conjecture 5

In the notation of [3], let $N \geq 2$, $1 \leq k < N$, and

$$g_{k,N}(y) = \log(q^k; q^N)_\infty, \quad f_{k,N}(y) = \log(e^{2\pi i k/N}; q)_\infty, \quad q = e^{2\pi i y}, \quad \Im y > 0.$$

Set

$$a = k/N, \quad h = -2\pi i Ny, \quad Q = e^{-4\pi^2/h} = e^{-2\pi i/(Ny)}. \quad (7.1)$$

Then $g_{k,N}(y) = p(a, h)$ and

$$\begin{aligned} L_+(a, Q) &= f_{k,N}(-1/(Ny)) - \log(1 - e^{2\pi i k/N}), \\ L_-(a, Q) &= f_{N-k,N}(-1/(Ny)) - \log(1 - e^{-2\pi i k/N}). \end{aligned}$$

Here the subtracted constants mean the analytically determined product factor at dual index zero. Writing the difference as $L_\pm$ avoids an uncontrolled multiple of $2\pi i$ from separately chosen logarithms.

The positive real Borel direction in the y variable rotates to the negative imaginary direction in the h variable, hence to our lower lateral continuation. The negative real y direction rotates to the upper one. Theorem 6.1 therefore gives

$$s_0(\tilde{g}_{k,N})(y) = g_{k,N}(y) - f_{k,N}(-1/(Ny)) + \log(1 - e^{2\pi i k/N}), \quad \Re y > 0, \quad (7.2)$$

$$s_\pi(\tilde{g}_{k,N})(y) = g_{k,N}(y) - f_{N-k,N}(-1/(Ny)) + \log(1 - e^{-2\pi i k/N}), \quad \Re y < 0. \quad (7.3)$$

Both domains are intersected with $\Im y > 0$. These are precisely (60a)–(60b), labeled Conjecture 5 in arXiv:2506.08265v2 [3]. Thus the Mellin calculation supplies the missing analytic identification in that statement. The excluded index $k = 0$ modulo N would make the isolated symbol $f_{0,N}$ singular; it must not be inserted into (7.2) as a finite logarithm.

7.2 Beyond Dirichlet characters

Theorem 7.1 (Antisymmetry is necessary and sufficient). *Fix $N \geq 2$ and arbitrary complex weights $w_1, \dots, w_{N-1}$. Put $w_0 = 0$ and*

$$P_w(h) = \sum_{k=1}^{N-1} w_k p(k/N, h), \quad \hat{P}_w = \sum_{k=1}^{N-1} w_k \hat{p}(k/N, h).$$

Then

$$P_w = \mathcal{M}\hat{P}_w \text{ identically} \iff w_k = -w_{N-k} \quad (1 \leq k < N). \quad (7.4)$$

If the equality fails, its first nonzero dual coefficient occurs at some power Q^n with $1 \leq n \leq N$.

Proof. By Theorem 6.1, the defect is

$$C_w(Q) = P_w - \mathcal{M}\hat{P}_w = - \sum_{m \geq 1} \frac{c_w(m)}{m} \frac{Q^m}{1 - Q^m}, \quad c_w(m) = \sum_{k=1}^{N-1} w_k \cos(2\pi k m/N). \quad (7.5)$$

Antisymmetry of the weights makes each $c_w(m)$ zero. Conversely, if $C_w = 0$, its coefficient at Q^n gives $\sum_{d|n} c_w(d)/d = 0$. Möbius inversion yields

$$c_w(n) = -n \sum_{d|n} \mu(n/d) [Q^d] C_w(Q) = 0. \quad (7.6)$$

The sequence $c_w(n)$ is the discrete Fourier transform of the even part $(w_k + w_{-k})/2$ of the cyclic weight vector. Invertibility of the finite Fourier transform implies that this even part is zero.

If it is nonzero, some $c_w(n)$ with $1 \leq n \leq N$ is nonzero by periodicity. Choose the least such positive n . All proper-divisor terms in the coefficient at Q^n vanish, leaving $-c_w(n)/n \neq 0$. This proves the finite detection bound. $\square$

A nonzero Dirichlet character satisfies $\chi(-k) = \chi(-1)\chi(k)$. For $N \geq 2$ its value at the zero residue is zero. Hence (7.4) says exactly that reconstruction holds if and only if the character is odd. Together with (7.1), this proves the statement labeled Conjecture 1 in [3]. Primitivity is not needed for this summability criterion.

7.3 Examples and what the theorem does not say

For $N = 3$, weights $(1, -1)$ give exact median reconstruction, while $(1, 1)$ give a nonzero even defect. For $N = 4$, antisymmetry requires $w_1 = -w_3$ and $w_2 = 0$. More generally the admissible weight space has dimension $\lfloor (N-1)/2 \rfloor$ over $\mathbb{C}$.

The result classifies reconstruction of these *linear logarithmic* combinations. Exponentiating an arithmetic mean or averaging inverses would be a different operation. Theorem 7.1 neither constructs an algebra-compatible median for all transseries nor settles the broader modular-resurgence conjectures in [3].

8 An exact q -gamma completion and its moving gamma core

8.1 The regularized observable

For $q = e^{-h}$, define the standard q -gamma function [10] by

$$\Gamma_q(a) = \frac{(q; q)_\infty (1-q)^{1-a}}{(q^a; q)_\infty}, \quad F(a, h) = \log \Gamma_{e^{-h}}(a). \quad (8.1)$$

For positive real a, h the logarithm is real, and local complex continuations use that branch. Cancellation of the essential and logarithmic terms in (5.2) gives the formal expansion

$$\widehat{F}(a, h) = \log \Gamma(a) + E_\Gamma(a, h) + \widehat{\Phi}_a(h), \quad (8.2)$$

$$E_\Gamma(a, h) = (1-a) \operatorname{Log} \frac{1-e^{-h}}{h} + \frac{B_2(1) - B_2(a)}{4} h. \quad (8.3)$$

The function E_Γ is convergent near $h = 0$; the generally divergent part is entirely in $\widehat{\Phi}_a$.

Theorem 8.1 (Exact gamma completion). *Let $B_\pm(a, h) = \mathcal{S}_\pm \widehat{F}(a, h)$ and $H(a, h) = (B_+(a, h) + B_-(a, h))/2$. In the local domains just specified,*

$$F = B_+ + \log(Q; Q)_\infty - L_-(a, Q) \quad (8.4)$$

$$= B_- + \log(Q; Q)_\infty - L_+(a, Q), \quad (8.5)$$

$$F = H + D(a, Q), \quad (8.6)$$

where

$$\begin{aligned} D(a, Q) &= \log(Q; Q)_\infty - \frac{L_+(a, Q) + L_-(a, Q)}{2} \\ &= \sum_{m \geq 1} \frac{\cos(2\pi m a) - 1}{m} \frac{Q^m}{1 - Q^m} = \sum_{n \geq 1} \delta_n(a) Q^n, \end{aligned} \quad (8.7)$$

$$\delta_n(a) = \sum_{d|n} \frac{\cos(2\pi d a) - 1}{d}. \quad (8.8)$$

The lateral jump is

$$B_+ - B_- = J(a, Q) = L_-(a, Q) - L_+(a, Q). \quad (8.9)$$

Proof. Take the limit $a \rightarrow 1$ in Theorem 6.1. The tail vanishes there, because $B_{2k+1}(1) = 0$ for $k \geq 1$, and $L_\pm(1, Q) = \log(Q; Q)_\infty$. This gives $p(1, h) = E(1, h) + \log(Q; Q)_\infty$. Insert this and (6.2)–(6.3) into $F = p(1, h) + (1-a) \log(1 - e^{-h}) - p(a, h)$. The core difference is (8.3); averaging and subtracting give the remaining identities.

For parameters above 1, use

$$F(a+1, h) - F(a, h) = \operatorname{Log} \frac{1 - e^{-ah}}{1 - e^{-h}}. \quad (8.10)$$

Its nonconstant small- h expansion is convergent locally. Both D and J are periodic in a , so the completion identities persist under the shift. Complex neighborhoods follow by analytic continuation and the parameter-holomorphic Borel kernel. $\square$

8.2 A completely invisible nonperturbative term

At $a = 1/2$, all the Bernoulli polynomials $B_{2k+1}(a)$ in (5.3) vanish. Thus the algebraic tail is zero and so is its Stokes jump. Nevertheless,

$$D(1/2, Q) = \log(Q; Q)_\infty - \log(-Q; Q)_\infty = -2Q - 2Q^2 + O(Q^3) \neq 0. \quad (8.11)$$

Here $\hat{F}$ is convergent near $h = 0$, but its sum is not the physical F : an essential flat term remains. This explicit example shows that neither an all-orders algebraic expansion nor its complete Stokes discontinuity determines the original function without additional normalization data.

For a positive integer a , in contrast, $D = J = 0$ and $\Gamma_q(a) = \prod_{j=1}^{a-1} (1 - q^j)/(1 - q)$ is a finite product. The difference between half-integers and integers is therefore visible in the flat completion even when both have zero Stokes jump.

8.3 Algebraic coefficients and parameter uniformity

Expanding (8.2) gives

$$F(a, h) = \log \Gamma(a) + f_1(a)h + f_2(a)h^2 + O(h^4), \quad (8.12)$$

with

$$f_1(a) = -\frac{(a-1)(a-2)}{4}, \quad f_2(a) = \frac{2a^3 - 3a^2 - 5a + 6}{144}. \quad (8.13)$$

There is no cubic term; after the linear term the formal expansion has only even powers. Formula (8.12) is uniform on compact complex neighborhoods of a positive real parameter, on smaller h -sectors. The same holds after any fixed number of a -derivatives. One proof uses the parameter-holomorphic Borel kernel, a uniform Laplace estimate, and Cauchy's estimate in a . The exact completion D is exponentially small on these compact sets.

At $a = 3$, the finite formula $\Gamma_q(3) = 1 + q$ gives $\log(1 + e^{-h}) = \log 2 - h/2 + h^2/8 + O(h^4)$, agreeing with (8.13). Such checks audit the normalization but do not replace the Mellin proof.

8.4 Explicit flat-tail bounds

If $|\Im a| \leq M$ and $\rho = |Q|e^{2\pi M} < 1$, then

$$|D(a, Q)| \leq \frac{-2 \log(1 - \rho)}{1 - |Q|}. \quad (8.14)$$

Indeed, bound $|\cos(2\pi ma) - 1|$ by $2e^{2\pi m M}$ in (8.7). An action-coefficient bound is $|\delta_n(a)| \leq 2(n+1)e^{2\pi n M}$, so truncating after Q^K gives

$$\left| D - \sum_{n=1}^K \delta_n(a) Q^n \right| \leq 2\rho^{K+1} \left(\frac{K+2}{1-\rho} + \frac{\rho}{(1-\rho)^2} \right). \quad (8.15)$$

The same type of estimate applies to J , and parameter derivatives follow by Cauchy's estimate on a slightly larger a -neighborhood. These bounds can be inserted directly into the inverse certificates; the numerical program is not needed to justify them.

9 Regular inverses in the gamma parameter and in h

9.1 Reversing the gamma parameter

Let a_0 be a positive real number with $\psi(a_0) \neq 0$, and fix $y = \log \Gamma(a_0)$ on a local complex target neighborhood. The inverse of $F(a, h) = y$ has the all-orders algebraic expansion

$$a(y, h) = a_0 - \frac{f_1(a_0)}{\psi(a_0)}h + \left(-\frac{f_2(a_0)}{\psi(a_0)} + \frac{f_1(a_0)f_1'(a_0)}{\psi(a_0)^2} - \frac{\psi_1(a_0)f_1(a_0)^2}{2\psi(a_0)^3} \right) h^2 + O(h^3), \quad (9.1)$$

where ψ_j denotes the order- j polygamma function, so $\psi_1 = \psi'$ and $\psi_2 = \psi''$. Coefficients at every order follow uniquely by substituting into (8.2). Uniform forward remainders and Lemma 3.2 show that this formal series is the asymptotic expansion of the analytic local inverse. In a chosen summable algebra, regular implicit closure supplies the corresponding lateral inverse [6]; the physical correction must still be added.

For that correction, let $a_H(y, h)$ be the actual inverse of the median-forward function H , and work on a compact regular chart where $|H_a(a_H, h)|$ stays positive. Theorem 8.1 and analytic inversion in the independent variable Q give

$$a_F - a_H = \frac{1 - \cos(2\pi a_H)}{H_a(a_H, h)}Q + O(Q^2). \quad (9.2)$$

The coefficient is an actual function of h . Replacing it by its leading algebraic value generally changes the error to $O(hQ)$, which is much larger than $O(Q^2)$. This is why full coefficient blocks are retained in an exponentially accurate inverse.

Similarly, if $G_{\pm}$ are regular inverses of $B_{\pm}$, then

$$G_+ - G_- = -\frac{2i \sin(2\pi G_-)}{(B_-)_a(G_-, h)}Q + O(Q^2), \quad (9.3)$$

and Theorem 3.1 supplies every higher jump term. The physical correction in (9.2) is not the same as the Stokes jump in (9.3).

9.2 Reversing the moving factorial with respect to h

Fix $a \in (0, 1)$ and put

$$A_0 = \pi^2/6, \quad \beta = \frac{1}{2} - a, \quad C(a) = \frac{\sqrt{2\pi}}{\Gamma(a)}, \quad x = A_0/h. \quad (9.4)$$

Given a small nonzero value $P = \exp p(a, h)$, choose the logarithmic sheet

$$\ell = \text{Log}(C(a)/P) + \beta \text{Log } A_0.$$

The physical inverse equation is

$$\ell = x + \beta \text{Log } x - \mathcal{R}(a, A_0/x) - C_q(a, e^{-24x}), \quad (9.5)$$

where

$$\mathcal{R}(a, h) = \frac{B_2(a)}{4}h - \mathcal{M}\hat{\Phi}_a(h), \quad C_q(a, Q) = \frac{L_+(a, Q) + L_-(a, Q)}{2}.$$

The exponent 24 is the exact action ratio $4\pi^2/A_0$. This is a logarithmic inverse chart on a cover, not a single-valued inverse on the punctured P -plane.

The elementary core $x + \beta \operatorname{Log} x = \ell$ has the selected solution $x \sim \ell$. For $\beta \neq 0$, it can be written

$$x_0 = \beta W\left(\beta^{-1}e^{\ell/\beta}\right), \quad (9.6)$$

where the Lambert branch is the one satisfying that asymptotic relation. For real large ℓ , it is W_0 when $\beta > 0$ and W_{-1} when $\beta < 0$. Complex continuation follows the chosen inverse chart rather than a global fixed branch label. At $\beta = 0$, $x_0 = \ell$.

Set

$$r_1 = \frac{A_0 B_2(a)}{4}, \quad r_2 = -\frac{A_0^2 B_3(a)}{72}, \quad L = \operatorname{Log} \ell.$$

Formal reversion of (9.5) through the algebraic orders gives

$$\begin{aligned} x \sim \ell - \beta L + \frac{\beta^2 L + r_1}{\ell} \\ + \frac{\frac{1}{2}\beta^3 L^2 - \beta^3 L + \beta r_1 L - \beta r_1 + r_2}{\ell^2} + \dots, \end{aligned} \quad (9.7)$$

$$h \sim \frac{A_0}{\ell} + \frac{A_0 \beta L}{\ell^2} + \frac{A_0(\beta^2 L^2 - \beta^2 L - r_1)}{\ell^3} + \dots. \quad (9.8)$$

Substituting successively determines every polynomial in $\operatorname{Log} \ell$. The coefficient of the next unknown is one in the normalized x equation. Finite-order sectorial remainders and the disk certificate prove the corresponding asymptotic inverse statements. These expansions are generally not convergent in the algebraic index.

9.3 A convergent lift for the complete exponential sectors

Proposition 9.1 (Action-24 inverse lift). *Let*

$$f_M(x) = x + \beta \operatorname{Log} x - \mathcal{R}(a, A_0/x), \quad f_M(X) = \ell,$$

with X on the selected large regular chart. There is a function $u = u(X, t)$ holomorphic for $|t| < t_0$, with $u(X, 0) = 0$, such that the physical inverse is

$$x = X + u(X, e^{-24X}), \quad h = \frac{A_0}{X + u(X, e^{-24X})}. \quad (9.9)$$

The radius t_0 may be chosen uniformly for large X on a closed subsector. Moreover,

$$u(X, t) = -\frac{\cos(2\pi a)}{f'_M(X)}t + O(t^2), \quad (9.10)$$

$$h - \frac{A_0}{X} = \frac{A_0 \cos(2\pi a)}{X^2 f'_M(X)}e^{-24X} + O\left(X^{-2}e^{-48X}\right). \quad (9.11)$$

All exponential-sector coefficients are obtainable by analytic implicit reversion, with geometric tails in t .

Proof. Substitute $x = X + u$ into (9.5). Its equation is

$$f_M(X + u) - f_M(X) - C_q(a, te^{-24u}) = 0. \quad (9.12)$$

For u in a fixed small disk, this is holomorphic in u, t . At $(u, t) = (0, 0)$ its u -derivative is $f'_M(X) = 1 + O(X^{-1})$, uniformly on the chosen subsector. All higher local bounds are uniform there, so the analytic implicit construction has a uniform t -radius. The coefficient of Q in C_q is $-\cos(2\pi a)$, proving (9.10). Expanding $A_0/(X + u)$ proves (9.11). Cauchy's estimate yields geometric remainders for the convergent lifted series. $\square$

The factor e^{-24u} in (9.12) is important: the exponential action is evaluated at the *unknown* inverse point. Freezing it at X beyond first order would miss action-feedback terms. The coordinate $x = A_0/h$ keeps this feedback uniformly analytic even though differentiation of $e^{-A/h}$ in h produces powers of h^{-2} .

At $a = 1/4$ or $3/4$, the action-24 coefficient vanishes. The first coefficient of C_q is then $Q^2/2$, so the first physical inverse correction has action 48 in X and equals $-A_0 e^{-48X}/(2X^2 f'_M(X))$ to leading sector order. This is an amplitude cancellation, not a disappearance of the dual scale itself.

10 The natural q -gamma minimum and half-action inverse roots

10.1 Existence and the algebraic moving center

For real $h > 0$, differentiation of the convergent logarithmic product gives

$$F_a(a, h) = -\log(1 - e^{-h}) - h \sum_{n \geq 0} \frac{1}{e^{h(n+a)} - 1}, \quad (10.1)$$

$$F_{aa}(a, h) = h^2 \sum_{n \geq 0} \frac{e^{h(n+a)}}{(e^{h(n+a)} - 1)^2} > 0, \quad a > 0. \quad (10.2)$$

The first derivative tends to $-\infty$ at $a = 0$ and to $-\log(1 - e^{-h}) > 0$ at infinity. Thus the physical function has exactly one positive critical point, a strict global minimum.

Let a_* be the unique positive zero of $\psi(a)$. Then

$$1 < a_* < \frac{3}{2}, \quad a_* = 1.4616321449683623412626595423 \dots \quad (10.3)$$

Uniqueness follows from $\psi_1(a) > 0$; the endpoint signs follow from $\psi(1) = -\gamma$ and $\psi(3/2) = 2 - \gamma - 2 \log 2$ [11]. The elementary bounds $\gamma < 3/5$ and $\log 2 < 7/10$ make the latter strictly positive. The accompanying exact checks verify these bounds using finite rational sums; see Section 13. In particular, neither $\sin(2\pi a_*)$ nor $1 - \cos(2\pi a_*)$ vanishes.

For small h on a sector, let $c(h)$ denote the critical point of the *median-forward* function H near a_* , and put

$$v(h) = H(c(h), h), \quad \alpha(h) = \frac{1}{2} H_{aa}(c(h), h). \quad (10.4)$$

Uniform parameter asymptotics and the analytic implicit theorem give this center, with $\alpha(h) \rightarrow \psi_1(a_*)/2 \neq 0$. It is not defined as an arithmetic median of critical points of $B_\pm$. That different operation would encounter Proposition 3.3.

Writing $c = a_* + c_1 h + c_2 h^2 + O(h^3)$, we obtain

$$c_1 = \frac{2a_* - 3}{4\psi_1(a_*)}, \quad (10.5)$$

$$c_2 = -\frac{\frac{1}{2}\psi_2(a_*)c_1^2 - \frac{1}{2}c_1 + f'_2(a_*)}{\psi_1(a_*)}, \quad (10.6)$$

$$v(h) = \log \Gamma(a_*) + f_1(a_*)h + \left(f_2(a_*) - \frac{f'_1(a_*)^2}{2\psi_1(a_*)} \right) h^2 + O(h^3). \quad (10.7)$$

These formulas follow by expanding $H_a(c, h) = 0$ and $H(c, h)$. The same recursive calculation gives all algebraic orders. The full functions $c(h), v(h), \alpha(h)$, rather than finite truncations of them, are used in the exponential statements below.

10.2 The physical critical value is different

Write $c_F(h)$ and $v_F(h)$ for the actual positive critical point and value of F . Since $F = H + \sum_{n \geq 1} \delta_n(a)Q^n$, equations (4.7)–(4.8) give

$$c_F = c - \frac{\delta'_1(c)}{H_{aa}(c, h)}Q + O(Q^2) = c + \frac{2\pi \sin(2\pi c)}{H_{aa}(c, h)}Q + O(Q^2), \quad (10.8)$$

$$v_F = v + \delta_1(c)Q + \left(\delta_2(c) - \frac{\delta'_1(c)^2}{2H_{aa}(c, h)} \right) Q^2 + O(Q^3). \quad (10.9)$$

Because $\delta_1(c) = \cos(2\pi c) - 1 < 0$ for small positive h , the physical minimum lies below the median-forward one. The error in its value is exponentially small, but it is the leading datum for the inverse in this target window.

Theorem 10.1 (Two real inverse roots with half the forward action). *For all sufficiently small positive h , the equation*

$$F(a, h) = v(h) \quad (10.10)$$

has exactly two positive real roots. They have the convergent lifted expansions

$$a_{\pm}(h) = c(h) \pm \sqrt{\frac{1 - \cos(2\pi c(h))}{\alpha(h)}} e^{-2\pi^2/h} + O(e^{-4\pi^2/h}). \quad (10.11)$$

The coefficient tends to

$$C_* = \sqrt{\frac{2(1 - \cos(2\pi a_*))}{\psi_1(a_*)}} = 2.0183815070093546945009855024 \dots \quad (10.12)$$

More generally, for $y = v(h) + QY$, every branch has an all-orders expansion in $Q^{1/2}$ on compact complex Y -patches avoiding $Y = \delta_1(c(h))$ uniformly.

Proof. Use the independent-parameter family $\mathcal{F}(a, h, t) = H(a, h) + D(a, t)$. It is uniformly holomorphic near $(a, t) = (c(h), 0)$, and $\mathcal{F}_{aa}(c, h, 0) = 2\alpha(h) \neq 0$. Theorem 4.1 with $r = 2$, $t = Q$, and $Y = 0$ yields (10.11). For positive h , the square root has a positive real argument, so both local branches are real and distinct. Furthermore, $F(c, h) = v + D(c, Q) < v$: every real term in (8.7) is nonpositive, with at least one strictly negative. Strict convexity (10.2), together with divergence of F at both positive endpoints, shows that these are exactly the two positive roots. The limiting coefficient follows from $c \rightarrow a_*$ and $2\alpha \rightarrow \psi_1(a_*)$. The all-orders conclusion is the convergent implicit expansion of Theorem 4.1. $\square$

The coefficient in (10.11) may be replaced by C_* only at the cost of an $O(he^{-2\pi^2/h})$ error. That error is much larger than $O(e^{-4\pi^2/h})$; the stronger displayed remainder requires retaining the moving coefficient.

10.3 The first three sectors explicitly

In (4.9)–(4.11), substitute

$$\alpha_j = \frac{\partial_a^j H(c, h)}{j!}, \quad d_0 = \delta_1(c), \quad d_1 = \delta'_1(c), \quad d_2 = \frac{1}{2}\delta''_1(c), \quad e_0 = \delta_2(c).$$

Then

$$a = c + u_0Q^{1/2} + u_1Q + u_2Q^{3/2} + O(Q^2), \quad u_0^2 = \frac{Y - \delta_1(c)}{\alpha_2}. \quad (10.13)$$

In particular, $\delta'_1 = -2\pi \sin(2\pi a)$, $\delta''_1 = -4\pi^2 \cos(2\pi a)$, and

$$\delta_2 = \cos(2\pi a) - 1 + \frac{\cos(4\pi a) - 1}{2}.$$

The two signs of u_0 give the two sheets. Their common Q coefficient u_1 need not be the physical critical-point shift: asymmetry of the cubic forward term also shifts the midpoint of the two roots.

The moving-fold representation (4.6), applied to $\mathcal{F}$, is the correct expression when the target approaches the physical branch point $v_F(h)$. Expansion (10.13) alone is not uniform at that additional coalescence.

10.4 A genuine critical Stokes comparison

The preceding real example compares a physical completion with the median-forward sum. To see the same action division in an actual lateral Stokes comparison, let $c_-(h)$ be the critical point of B_- near a_* , put $v_- = B_-(c_-, h)$, and set $\alpha_- = (B_-)_{aa}(c_-, h)/2$. Since $B_+ = B_- + J(a, Q)$ and $J = 2i \sin(2\pi a)Q + O(Q^2)$, the upper lateral function has, at the lower lateral critical target, the two local inverse values

$$G_{+,\pm}(v_-, h) = c_-(h) \pm \sqrt{-\frac{2i \sin(2\pi c_-(h))}{\alpha_-(h)}} Q^{1/2} + O(Q). \quad (10.14)$$

The inverse of B_- itself has a double value c_- at that target. The coefficient in (10.14) is nonzero for small h , since $a_* \in (1, 3/2)$. On neighboring target patches the full comparison is (4.4), not a discontinuous rule for choosing the sign of a square root.

The physical q -gamma function is analytic across the positive h axis. Its completion terms in (8.4) and (8.5) cancel the jump of the perturbative lateral sums. Equation (10.14) compares those lateral sums; it does not claim that the physical function itself has a discontinuity.

11 A convergent reflection model with no Stokes jump

A particularly transparent critical example is obtained without any divergent algebraic block. Consider

$$R(a, h) = \log(\Gamma_{e^{-h}}(a)\Gamma_{e^{-h}}(1-a)), \quad 0 < a < 1. \quad (11.1)$$

The identity $B_{2k+1}(1-a) = -B_{2k+1}(a)$ cancels the entire tail. The ordinary gamma reflection formula [11] gives the exact convergent core

$$R_0(a, h) = \log \frac{\pi}{\sin(\pi a)} + \text{Log} \frac{1 - e^{-h}}{h} + \frac{a(1-a)}{2}h. \quad (11.2)$$

The completion and jump are

$$R(a, h) = R_0(a, h) + 2D(a, Q), \quad J(a, Q) + J(1-a, Q) = 0. \quad (11.3)$$

Thus R_0 is genuinely convergent near $h = 0$; it has no lateral Borel ambiguity at all. Nevertheless, the flat term $2D$ remains.

Theorem 11.1 (Explicit reflection inverse transseries). *The equation*

$$R(a, h) = R_0(1/2, h) \quad (11.4)$$

has two inverse branches near $a = 1/2$ of the form

$$a_{\pm} = \frac{1}{2} \pm Q^{1/2}U(h, Q), \quad (11.5)$$

where U is jointly holomorphic near $(h, Q) = (0, 0)$, and

$$U(h, 0) = \sqrt{\frac{8}{\pi^2 - h}}. \quad (11.6)$$

The square root is positive at $h = 0$. In particular, the complete lifted transseries converges in (h, Q) , and only odd powers of $Q^{1/2}$ occur in the displacement from $1/2$.

Proof. Both $R_0(1/2 + z, h) - R_0(1/2, h)$ and $D(1/2 + z, Q)$ are even holomorphic functions of z , so they are holomorphic in $w = z^2$. Their equation is

$$\frac{\pi^2 - h}{2}w + O(w^2) - 4Q + O(Qw, Q^2) = 0. \quad (11.7)$$

At $(w, h, Q) = (0, 0, 0)$ the w -derivative is $\pi^2/2 \neq 0$. The analytic implicit theorem gives a unique holomorphic $w(h, Q)$ vanishing at $Q = 0$. It factors as $w = QV(h, Q)$ with $V(h, 0) = 8/(\pi^2 - h)$, which is nonzero near $h = 0$. Choose $U = \sqrt{V}$. Then $z = \pm\sqrt{Q}U$ proves the theorem. $\square$

This example isolates two phenomena that are often conflated. A Stokes jump is absent, but a flat completion is present; that completion still creates half-action inverse roots. It also shows that divergent perturbative series are not necessary for exponentially amplified critical inverse errors.

11.1 Explicit higher sectors

Write $w = b_1Q + b_2Q^2 + b_3Q^3 + \dots$ and set

$$\alpha = \frac{\pi^2 - h}{2}, \quad \beta_4 = \frac{\pi^4}{12}, \quad \beta_6 = \frac{\pi^6}{45}.$$

Expansion of (11.7) gives

$$b_1 = \frac{4}{\alpha}, \quad (11.8)$$

$$b_2 = \frac{4 - \beta_4 b_1^2 - 4\pi^2 b_1}{\alpha}, \quad (11.9)$$

$$b_3 = \frac{\frac{16}{3} - 2\beta_4 b_1 b_2 - \beta_6 b_1^3 - 4\pi^2 b_2 + \frac{4}{3}\pi^4 b_1^2 + 4\pi^2 b_1}{\alpha}. \quad (11.10)$$

Consequently,

$$a_{\pm} = \frac{1}{2} \pm \sqrt{b_1}Q^{1/2} \left[1 + \frac{b_2}{2b_1}Q + \left(\frac{b_3}{2b_1} - \frac{b_2^2}{8b_1^2} \right) Q^2 + O(Q^3) \right]. \quad (11.11)$$

Every coefficient is an analytic function of h near zero. At $h = 0$, $b_1 = 8/\pi^2$ and $b_2/(2b_1) = -25/6$. These are coefficients of the lift, not a substitution of $h = 0$ into the original essential variable $Q = e^{-4\pi^2/h}$.

For $0 < h < \pi^2$, R_0 is strictly convex on $(0, 1)$ because its second derivative is $\pi^2 \csc^2(\pi a) - h > 0$. The physical R is strictly convex there by (10.2). Both are symmetric about $1/2$, while the physical minimum is lower by

$$2D(1/2, Q) = -4Q - 4Q^2 + O(Q^3).$$

Hence the two real roots of (11.4) are not an artifact of complex branch notation. The omitted flat term changes the actual inverse multiplicity at the chosen real target.

12 Accuracy, action budgets, and compatibility of charts

12.1 Critical residuals are Hölder, not Lipschitz

Lemma 12.1 (Critical root-cluster certificate). *Suppose $F_0(c+z) - v = z^r A(z)$ on $|z| \leq R_0$, with $|A(z)| \geq a_0 > 0$. If $|F(c+z) - F_0(c+z)| \leq \varepsilon$ there, and $R = (2\varepsilon/a_0)^{1/r} < R_0$, then $F(x) = v$ has exactly r roots, counting multiplicity, in $|x - c| < R$.*

Proof. On $|z| = R$, $|F_0(c+z) - v| \geq a_0 R^r = 2\varepsilon$. Rouché's theorem compares $F - v$ with $z^r A(z)$, which has exactly r roots in that disk. $\square$

This gives root existence and the scale $\varepsilon^{1/r}$. It does not by itself label the roots. After the scaled roots separate, the ordinary disk certificate applies in the variable u from Theorem 4.1 and supplies branchwise errors.

An inverse tolerance on the scale $e^{-B/h}$ at an order- r critical point generally demands a forward tolerance on the scale $e^{-rB/h}$. Every fixed algebraic remainder $O(h^M)$ is much larger than the Q -sized target window in the q -gamma example. Substituting a few more algebraic terms is therefore not a substitute for identifying the flat completion. Even an optimally truncated approximation must carry a sufficiently accurate remainder description to determine the leading root coefficient in that window.

12.2 More general monomials and a small curvature

Theorem 4.1 is an analytic parameter theorem and can be applied to any chosen small holomorphic monomial

$$t(h) = h^\beta (\text{Log } h)^m e^{-A(h)}$$

on a fixed cover. The inverse monomial is $t(h)^{1/r}$ when its leading coefficient is nonzero and the uniform hypotheses hold. Thus algebraic, logarithmic, and nonlinear exponential actions can all be transported by the same local construction.

The nonvanishing curvature hypothesis matters. For example,

$$F(x, h) = e^{-B/h} x^r + e^{-A/h}, \quad A > B > 0,$$

at target zero has roots of size $e^{-(A-B)/(rh)}$, not $e^{-A/(rh)}$. Dividing by the small leading coefficient changes the effective action before ramification. An analogous normalization is valid for more general cores only when the normalized higher derivatives remain controlled on the required shrinking chart.

12.3 Three distinct sources of multivaluedness

There are three independent continuation operations in the examples. First, changing the logarithm of P changes ℓ by a multiple of $2\pi i$ in (9.5). Second, continuing around a critical target value permutes the square-root or higher Puiseux sheets. Third, changing a Borel summation direction across a singular ray changes a sectorial block by its Stokes map. Each has a different base variable and a different transition law.

A useful analytic atlas therefore records a source sector, a target patch, all logarithm branches, the selected inverse sheet, and the summation/completion convention. On overlaps, equations (3.3)–(3.4) give the regular transition identities; the moving fold chart supplies the branched extension. This is a local compatibility framework, not a claim of a single globally ordered transseries field on the entire complex plane.

12.4 What is and is not covered at other q -critical points

The explicit proofs concern $q \rightarrow 1$ with $z = q^a \rightarrow 1$. For a primitive k th root ζ , the exact residue-class splitting

$$(e^{-ah}; \zeta e^{-h})_\infty = \prod_{j=0}^{k-1} (\zeta^j e^{-(a+j)h}; e^{-kh})_\infty \quad (12.1)$$

separates a moving gamma block from the nontrivial cyclotomic blocks. The factor $j = 0$ is covered here after $h \mapsto kh$, $a \mapsto a/k$. The other factors require additional cyclotomic completion data and may interfere after differentiation. We do not transfer the action $4\pi^2$ or the median criterion unchanged to that entire product. The preceding repository q -cusp report treats a different modular Euler-quotient subclass [2].

13 Reproducible verification and numerical diagnostics

13.1 Exact checks

The accompanying `verify.py` uses SymPy for finite exact identities and mpmath for high-precision diagnostics. It requires neither repository files nor network access. The command

```
python verify.py -output results.json
```

reproduces the recorded tests. The supplied run uses 110 decimal digits and passed 144 exact assertions. These include the Bernoulli generating kernel through degree 12, the first gamma coefficients, finite-product q -gamma expansions at integers 2 through 8 through degree 6, the second-order inverse response, the fold coefficients (4.10)–(4.11), the polynomial–logarithmic inverse, the three reflection sectors, and finite Möbius inversions.

Two exact rational comparisons also justify the constants used to locate a_* in (10.3). Since $\gamma < H_{30} - \log 30$ and

$$\log 30 = 2 \sum_{j \geq 0} \frac{(29/31)^{2j+1}}{2j+1},$$

retaining 100 positive terms gives an exact lower bound for $\log 30$ that makes $H_{30} - \log 30 < 3/5$. Also the first six terms of $\exp(7/10)$ already exceed 2, proving $\log 2 < 7/10$. These tests are finite rational calculations, not floating-point estimates of the endpoint signs.

13.2 A numerically independent completion check

To compare the physical product with the Borel calculation, the program evaluates $p(a, h)$ directly using the convergent q -product. It evaluates $\mathcal{M}\hat{\Phi}_a$ separately by the kernel series (6.18). The latter is accelerated by subtracting its first ten algebraic kernel terms and summing those subtractions exactly through the known Bernoulli coefficients:

$$\begin{aligned} \mathcal{M}\hat{\Phi}_a(h) &= \sum_{k=1}^K c_{2k}(a) h^{2k} \\ &\quad + \frac{2}{\pi} \sum_{n \geq 1} \sigma_n(a) \left[K(An/h) - \sum_{k=1}^K \frac{(2k-1)!}{(An/h)^{2k}} \right], \\ c_{2k}(a) &= \frac{B_{2k} B_{2k+1}(a)}{2k(2k+1)!}, \quad \sigma_n(a) = \sum_{d|n} \frac{\sin(2\pi da)}{d}. \end{aligned} \quad (13.1)$$

This identity follows by absolute convergence after subtraction and the Bernoulli Fourier identities. The finite numerical cutoff is not outward-rounded or interval-certified. Comparing action cutoffs 60 and 120 is a diagnostic, not a certified tail enclosure.

a	h	Completion residual, cutoff 120	Change from cutoff 60
0.3	2	8.03×10^{-55}	1.55×10^{-48}
0.5	2	8.32×10^{-112}	5.07×10^{-159}
0.7	3	6.01×10^{-51}	1.16×10^{-44}

The tiny middle row is a degenerate exact-tail case: $\widehat{\Phi}_{1/2} = 0$. It is not evidence of superior numerical conditioning for general a . The residuals verify the chosen signs and normalizations to the displayed numerical accuracy. The proof of the infinite identity is the Mellin argument in Section 6.

13.3 Critical-root scaling

For the natural gamma minimum, the program constructs $H = F - D$ from independent physical and dual-product formulas, finds its critical point c , and solves the physical equation $F(a, h) = H(c, h)$. The root solve is performed in the scaled variable

$$u = \frac{a - c}{\sqrt{(1 - \cos(2\pi c))/\alpha} \sqrt{Q}},$$

not in the ill-conditioned unscaled variable. The predicted limiting values are $u = \pm 1$.

h	Q	Predicted positive displacement	$ u_+ - 1 $
2	2.67529×10^{-9}	1.79327×10^{-4}	2.292×10^{-4}
1	7.15717×10^{-18}	6.98150×10^{-9}	6.011×10^{-9}
0.5	5.12250×10^{-35}	1.63654×10^{-17}	1.122×10^{-17}

The two physical root residuals in these tests are below 2×10^{-111} at the working precision. This is a floating-point residual, not an interval-certified bound on the exact roots. The displacement is much larger than Q , and the normalized error decreases on the expected $\sqrt{Q}$ scale.

The reflection model is checked against the *actual* product $\Gamma_q(a)\Gamma_q(1-a)$ rather than only its completion formula. At $h = 1$, the predicted leading displacement is $2.54075862518576 \times 10^{-9}$. The observed coefficient $(u_+ - 1)/Q$ is $-4.7764454296641167\dots$, versus $b_2/(2b_1) = -4.7764454296641171\dots$. At $h = 2$, the corresponding values differ by about 2.23×10^{-7} , consistent with the next Q correction. Full values are retained in `results.json`.

13.4 What was not verified computationally

No computation proves the Mellin contour deformation, a resurgent closure theorem, the Banach-analytic parameter result, or a universal complex continuation theorem. Those are addressed by the conventional proofs and the stated classical dependencies. No Lean proof, interval root certificate, or independently refereed verification is claimed. The tests are intended to make sign, coefficient, and branch mistakes easier to detect and reproduce.

14 Further research questions and topics

The following questions are not asserted to be solved by this article. They are organized around specific obstructions rather than a claim of unrestricted inversion of every possible transseries.

14.1 Inverse Borel singularities and local monodromy

For the regular inverse of $a \mapsto \log \Gamma_q(a)$, determine the precise Borel singularities of each inverse coefficient block, not just the allowed additive action lattice. The forward kernel is meromorphic with explicitly known residues. Nonlinear inversion introduces convolutions and can change poles into logarithmic or more complicated singularities. A useful goal is an explicit bridge equation for the inverse in a selected target chart, together with its continuation when the target approaches a fold.

14.2 A full nonlinear median at a critical value

The arithmetic median used here is a linear logarithmic construction. Build an algebra-compatible transseries completion for the full inverse family, with its transseries parameters transformed by Stokes maps. Determine how its median convention interacts with the ramified coordinate $\sqrt{y - v(h)}$. Proposition 3.3 shows that averaging already-inverted lateral branches is not an acceptable shortcut.

14.3 Confluence with gamma poles and zeros

The present parameter domains remain away from the nonpositive integers. When a approaches a pole of $\Gamma(a)$ simultaneously with $h \rightarrow 0$, the gamma core itself becomes singular. Construct a uniform inverse chart for $a = -m + \lambda h^\nu$ and identify the correct gamma, logarithmic, and exponential scales. The local continuation of the Borel kernel is not enough: pole removal and inverse-sheet selection must be controlled together.

14.4 Cyclotomic completions at general roots of unity

Use (12.1) to extend the Mellin argument to the complete residue-class family. The Fourier functional equation is then replaced by a finite phase-coupled system. Determine its exact even completion, its lateral jumps, and the analogue of Theorem 7.1. One should expect the inverse critical geometry to depend on cancellations between cyclotomic blocks, not merely on their individual action sizes.

14.5 Higher critical points in natural gamma quotients

Study finite products

$$\prod_{j=1}^m \Gamma_q(\lambda_j a + \mu_j)^{r_j}$$

on domains where all limiting gamma arguments are regular. Their completion is an explicit linear combination of D , and their jump is an explicit combination of J . Find natural parameter families in which the first $r - 1$ derivatives of the limiting logarithm vanish, while the r th does not. Theorem 4.1 then predicts an A/r action, unless a completion jet cancels. Classify when those cancellations are forced by reflection or multiplication identities.

14.6 Joint action and algebraic optimal truncation

An inverse approximation has two distinct truncation indices: the order inside an algebraic block and the number of exponential blocks retained. Develop certified joint choices for a prescribed inverse tolerance, using the exact kernel and the critical root-cluster certificate. A particularly useful target is a rigorous error bound that remains uniform while $y - v_F(h)$ passes through the Q window and then through a smaller window where the two inverse roots are close to merging.

14.7 Countable actions beyond local finiteness

Proposition 2.2 gives a Banach-analytic inverse for normally summable perturbations. Determine which of these inverses also possess a canonical formal transseries representation when action values accumulate. A norm can make an analytic sum meaningful even when a usual coefficient-by-coefficient action filtration fails. The missing ingredient is a support-sensitive uniqueness theorem, not just another contraction estimate.

14.8 Global inverse atlases and growing cusp denominator

For q -families continued between rational boundary charts, construct source and target atlases that track logarithmic sheets, Stokes maps, and critical-value monodromy simultaneously. Then investigate joint limits in which a root-of-unity denominator grows with $1/h$. The local fixed-denominator identities do not automatically provide uniform separation of inverse sheets or a globally summable action set.

14.9 Basic hypergeometric sums and saddle interference

Extend the completed-product method to basic hypergeometric functions, Nahm sums, and products occurring in quantum-dilogarithm integrals. Their forward asymptotics may contain several saddles rather than one gamma core. Derivative cancellation between saddles can create inverse critical points on a different scale from the forward Stokes transition. A successful theorem should identify both the saddle completion and the inverse branch before claiming an error bound.

14.10 Formalization and certification

A practical formalization can proceed in layers. The finite Fourier and Möbius criterion, the inverse coefficient identities, and the rational sign bounds are algebraic starting points. The next layer is the filtered contraction and the analytic fold factorization. The Mellin–Borel theorem then needs library support for principal-value kernels, contour shifts, and uniform infinite sums. Separately, a complex-interval implementation could certify particular inverse roots using Lemmas 3.2 and 12.1 without first formalizing the full resurgence theory.

15 Conclusion

The missing information in a complex asymptotic inverse is often not another algebraic coefficient. It is the analytic completion and the location of the moving inverse branch point. For the moving q -Pochhammer family, the Mellin calculation separates the remainder into a sine part, which is the symmetric Borel sum, and a cosine part, which is an independent convergent dual product. This proves the two specified summability statements and makes their scope explicit.

Once the completed function is known, regular analytic inversion gives full exponential sectors, while critical scaling gives fractional actions. The q -gamma minimum exhibits both a genuine half-action Stokes comparison and a real physical inverse splitting. The reflection product gives an even cleaner limiting lesson: a convergent core and zero Stokes jump can coexist with a nonzero flat term that changes the inverse problem at leading exponential order.

The resulting calculus is local, branch-aware, and exact in its flat parameters. It does not eliminate the need to choose a summation class, a sector, or an inverse sheet; it makes those choices part of the mathematical statement.

A Proof dependencies and research-status audit

Result	Main ingredients	Boundary of the claim
Borel kernel and residues	Bernoulli generating functions; normally convergent meromorphic sums	The rational forward singularity data are already known; complex-parameter continuation uses the non-Fourier kernel.
Exact completion	Mellin inversion; Hurwitz and Riemann functional equations; principal-value kernel	First proved for $h > 0$, $0 < a < 1$; then analytic continuation in the specified domains. No residue-only argument is substituted for reconstruction.
Conjecture 5 comparison	Explicit rotation $h = -2\pi i N y$ and removal of the dual index-zero factor	$1 \leq k < N$, $\Im y > 0$, with the specified lateral logarithms. No value is assigned to $\log(1 - 1)$.
Weight criterion and Conjecture 1	Completion identity; divisor Möbius inversion; finite Fourier inversion	Arbitrary finite complex weights, including characters. This is a linear median statement.
Regular inversion	Classical observable Lagrange inversion; analytic implicit theorem; disk contraction	A selected regular chart. Summation compatibility requires a suitable summable algebra.
Critical action division	Ramified analytic blow-up and simple scaled roots	Compact target patches away from the scaled discriminant; nondegenerate normalized leading coefficient.
Moving-fold chart	Parametric analytic Morse factorization	Valid at the moving branch point on a square-root cover, not as a single-valued holomorphic inverse in the target.
Natural gamma fold	Exact completion; positive q -trigamma; uniform parameter asymptotics	Small h near the classical positive minimum. All exponential estimates keep the moving coefficient blocks.
Reflection inverse	Gamma reflection; Bernoulli parity; analytic IFT in z^2, h, Q	A genuinely convergent local lift. The physical variable relation is imposed only after constructing the lift.
Numerical results	Exact symbolic checks and 110-digit evaluation	Not interval certificates, independent referee reports, or proof-assistant verification.

Scope of the source audit

The repository comparison uses its transseries documentation and the identified prior report's fixed-parameter, jump-transport, and further research sections. It is not a statement-by-

statement audit of the full canonical volume. The cited public literature was checked on 4 October 2026. The inspected Fantini–Rella version explicitly labels the two summability statements as conjectures; the present proof targets their formulas rather than a broader slogan about modular resurgence.

The derived Mellin argument and applications have undergone an internal algebraic and analytic consistency review in preparing this package. The principal-value interchange, lateral orientation, dual product constant, complex-parameter continuation, and critical target exclusions are spelled out to facilitate external review. Such an internal review cannot certify first publication priority or replace an independent expert assessment.

B A compact formula and sign reference

The conventions most likely to cause a sign error are collected here. Let $\Phi_{\pm} = \mathcal{S}_{\pm} \widehat{\Phi}_a$, $L_{\pm} = \log(e^{\pm 2\pi i a} Q; Q)_{\infty}$, and $L_0 = \log(Q; Q)_{\infty}$. Then

$$\begin{aligned} J &= \Phi_+ - \Phi_- = L_- - L_+, & J &= 2i \sin(2\pi a) Q + O(Q^2), \\ \mathcal{S}_+ \widehat{p} &= E - \Phi_+ = p - L_-, & \mathcal{S}_- \widehat{p} &= E - \Phi_- = p - L_+, \\ p - \mathcal{M} \widehat{p} &= (L_+ + L_-)/2, & p - \mathcal{M} \widehat{p} &= -\cos(2\pi a) Q + O(Q^2), \\ F - B_+ &= L_0 - L_-, & F - B_- &= L_0 - L_+, \\ F - H &= D = L_0 - (L_+ + L_-)/2, & D &= (\cos(2\pi a) - 1) Q + O(Q^2). \end{aligned}$$

The contour difference “upper minus lower” is clockwise and equals $-2\pi i$ times the residue. This explains the positive sign in the first line despite the negative residue in (5.6).

For a regular inverse of $f + td$, the first response is $-td/f'$. For the h -inverse of the factorial, the first physical correction is therefore positive when $\cos(2\pi a) > 0$, as displayed in (9.11). For the gamma parameter inverse the first correction is $(1 - \cos(2\pi a_H))Q/H_a$, as in (9.2). At a fold these linear expressions are replaced by the ramified balance.

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